Shiv Nadar University
Monsoon 2020, PHY101, Mid-term

Time: 2 Hrs Full Marks: [25]

1) Refer to the figure 1, below where two blocks of unequal mass are connected by a string over a smooth pulley. The co-efficient of kinetic friction between the block of mass '2m' and the incline is μ_k. If $\theta = \theta_0$ is the angle of incline which allows the masses to move at a constant speed, show that [5]

$$(1 - 2\sin\theta_0) = 2\mu_k \cos\theta_0$$

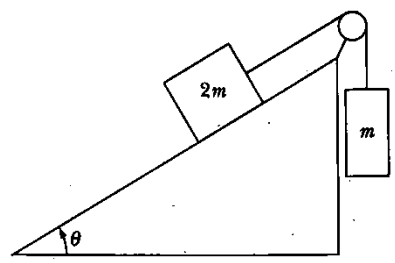

Figure 1

2) A ladder LN (shown in the figure 2) of length "a" rests against a vertical wall OL. The foot N of the ladder is pulled away with a constant speed v_0 along x-axis while it is always touching the ground.

(a) Show that the midpoint of the ladder (M) describes the arc of a circle of radius "$a/2$" with centre at O. [2]

(b) Find the velocity of the midpoint (M) of the ladder at the instant when N is at a distance b from the wall (b<a). [3]

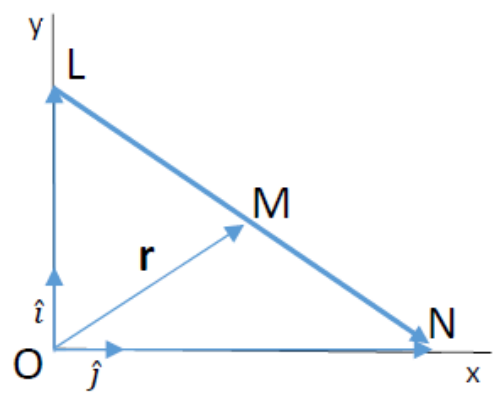

Figure 2

3) A particle of mass 2 kg is moving along x-axis with a potential energy given by

$$U(x) = a_1 x^2 - a_2 x^3 \text{ where } a_1 = 10 \text{ J}/m^2 \text{ and } a_2 = 2 \text{ J}/m^3$$

a) Find the stable equilibrium point of the motion. [2]

b) If the particle starts with zero speed at $x = a_1/a_2$, find its speed at the stable equilibrium point. [2]

4) A block mass 500 gm attached to a spring of spring constant 12.5 N/m undergoes a lightly damped oscillation with a frequency 0.2 % lower than its natural frequency.

a) What is the value of damping constant b (Consider damping force, $\vec{F}_{Damp} = -bv$, where v is the velocity)? [2]
b) How does the amplitude vary with time? [1]
c) Find the time in which amplitude decays ½ of its initial value? [1]
d) What would be the value of damping constant in the case of critical damping? [1]

5) A particle of mass m is constrained to move on a circular path of radius R, in a vertical plane as shown in the figure 3. A force $\vec{F}$ of constant magnitude F, is applied on particle toward the top point T of the circle. (Acceleration due to gravity is g.)

i) Find the work done by the force $\vec{F}$ in changing the angle $\angle$ BOA = θ from θ_1 to θ_2. [2]

ii) Find the work done by the weight in changing the angle $\angle$ BOA = θ from θ_1 to θ_2. [2]

iii) If the speed of the particle at angle θ_1 is v_0, what is the speed of particle at angle θ_2. [2]

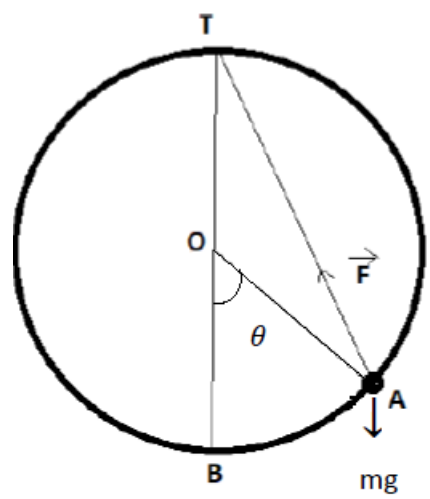

Figure 3